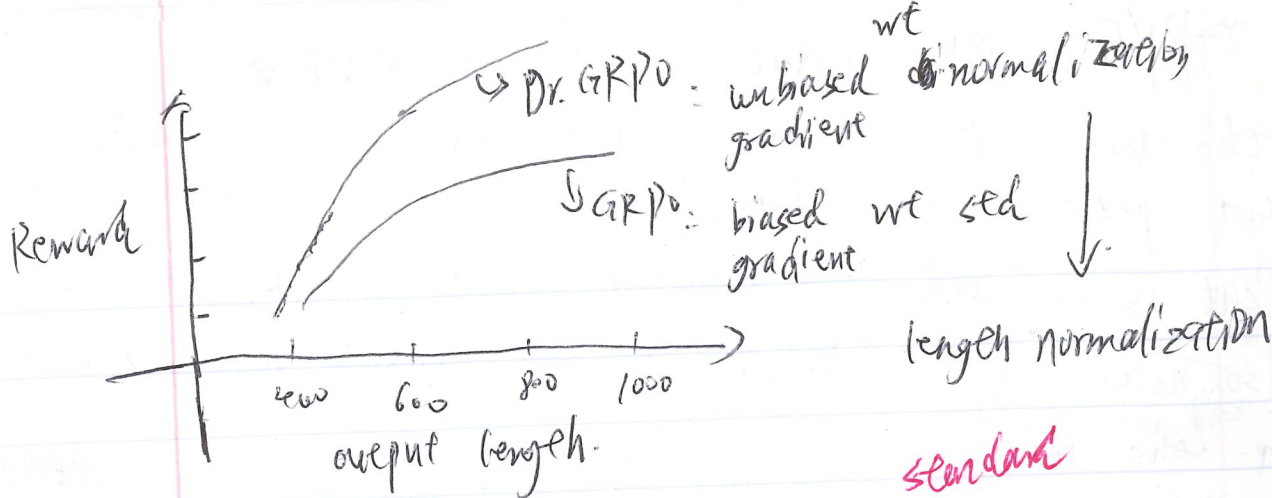




How is the biased gradient calculated in ~~GRPO~~ ^{standard} GRPO?

$$A_i = \frac{\overset{\text{current reward}}{r_i} - \overset{\text{mean reward}}{u}}{\text{std} + \epsilon} \cdot G_i + \tilde{A}_i$$

$\Rightarrow$ where std ~~relates~~ ^{compute over the group, including the current r_i} relates to current rewards r_i

Higher r_i : increase std , $\rightarrow$ reduce the scaled advantage A_i

Lower r_i : decrease std , $\rightarrow$ inflating $|A_i|$ for bad response

$\Rightarrow$ The unbiased $\bar{\pi}$ Dr. GRPO remove the problematic terms:

$$A_i = r_i - u \text{ (or minor adjustments to length term)}$$

Case studies:

[Jan, 22nd, 2025]

Deepseek-R1: Central to many recent RLVR efforts

Lots of interesting details

$\Rightarrow$ what is remarkable about R1?

- ①: performance exceeding OpenAI o1.
- ②: Open RL Recipe: pretty simple.
- ③: SFT insights (R1-zero/dsrl-R1) $\rightarrow$ ended speculations on necessity of MCTS/PRMs

GRPO: they build on the results of GRPO from Deepseek March

↓
dysk RI-zero
↓

deepseek-V3 → SFT-Reasoning and RL (GRPO)

↓
deepseek-RI-zero → ~~post-training~~ alignment: SFT / RLHF

⇒ controlled setting - RI zero.

see up: Rewards

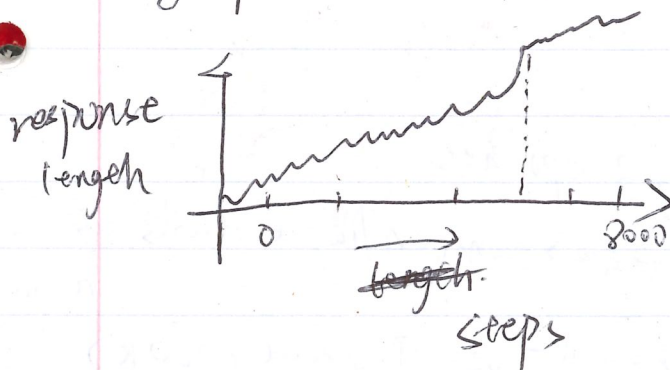
- accuracy rewards (is it correct?)
- Format rewards (use thinking tags)

Data: not public.

Base Model: Deepseek-V3

Results: match openAI-o1.

interesting phenomena: aha moment: emergent intelligence boost



↓ generate
⇒ longer CoTs during Training
/ well-formatted

⇒ pushing performance further - RI answers
RI vs RI-zero (duplicate, ill-formatted, mix-language, redundant, reply ✓)

key differences: ①: SFT initialization

②: language consistency reward for CoT.

③: non-verifiable rewards in stage 2.

Deepseek-V3 → Reasoning SFT → RL (GRPO) → SFT / RLHF
Reasoning ability alignment